

Week 7: SQL Query Implementation for E-Commerce Database

Scenario:

The e-commerce company has collected customer, product, order, and inventory data. The management team wants to retrieve useful information from the database for daily operations and basic business analysis.

Students will apply SQL queries to search, filter, sort, and generate reports from the existing E-Commerce Database.

Tasks to be Done:

1. Perform Basic SQL Queries

Concept:

SQL **SELECT** command is used to retrieve data from database tables.

- 1. Display all customer details.

```
SQL> SELECT * FROM Customer;
```

CUSTOMER_I	FIRST_NAME	LAST_NAME	EMAIL	PHONE	CREATED_A
CUST000001	Arjun	Kumar	arjun.kumar@gmail.com	9876543210	15-JAN-24
CUST000002	Priya	Sharma	priya.sharma@gmail.com	9123456780	28-MAR-24
CUST000003	Rahul	Menon	rahul.menon@gmail.com	9988776655	10-JUN-24
CUST000004	Ananya	Iyer	ananya.iyer@gmail.com	9876123450	22-AUG-24
CUST000005	Vikram	Reddy	vikram.reddy@gmail.com	9012345678	05-NOV-24
CUST000006	Meera	Krishnan	meera.krishnan@gmail.com	9345678120	18-JAN-25
CUST000007	Suresh	Nair	suresh.nair@gmail.com	9567891234	12-MAR-25
CUST000008	Kavya	Rajan	kavya.rajan@gmail.com	9789012345	25-MAY-25
CUST000009	Karthik	Subramanian	karthik.subramanian@gmail.com	8899776655	09-JUL-25
CUST000010	Divya	Pillai	divya.pillai@gmail.com	9098765432	30-SEP-25
CUST000011	Aditya	Verma	aditya.verma@gmail.com	9876501234	14-DEC-25
CUST000012	Sneha	Patel	sneha.patel@gmail.com	9765432109	20-FEB-26
CUST000013	Rohan	Bose	rohan.bose@gmail.com	9654321098	17-APR-26
CUST000014	Lakshmi	Venkatesan	lakshmi.venkatesan@gmail.com	9543210987	08-JUL-26
CUST000015	Nithin	Murugan	nithin.murugan@gmail.com	9432109876	03-OCT-26

15 rows selected.

2. Display all available products.

```
SQL> SELECT * FROM Product;
```

PRODUCT	PRODUCT_NAME	CATEGOR	PRICE
STOCK_QUANTITY			
P000001	Laptop 25	CAT001	55000
P000002	Wireless Mouse 100	CAT002	850
P000003	Mechanical Keyboard 60	CAT002	2500
PRODUCT PRODUCT_NAME CATEGOR PRICE			
STOCK_QUANTITY			
P000004	Bluetooth Speaker 45	CAT003	3200
P000005	Smartphone 35	CAT004	28000
P000006	USB Cable 150	CAT005	450
PRODUCT PRODUCT_NAME CATEGOR PRICE			
STOCK_QUANTITY			
P000007	Power Bank 75	CAT005	1800
P000008	Smart Watch 40	CAT006	6500
P000009	Headphones 55	CAT003	4200
PRODUCT PRODUCT_NAME CATEGOR PRICE			
STOCK_QUANTITY			
P000010	Tablet 30	CAT004	22000
P000011	Web Camera 50	CAT007	2800
P000012	External Hard Disk 35	CAT008	5200
PRODUCT PRODUCT_NAME CATEGOR PRICE			
STOCK_QUANTITY			
P000013	Gaming Monitor 20	CAT009	18500
P000014	Laptop Stand 80	CAT010	1500
P000015	Wireless Earbuds 65	CAT003	3500

15 rows selected.

3. Retrieve product names and prices only.

```
SQL> SELECT Product_Name, Price FROM Product;
```

PRODUCT_NAME	PRICE
Laptop	55000
Wireless Mouse	850
Mechanical Keyboard	2500
Bluetooth Speaker	3200
Smartphone	28000
USB Cable	450
Power Bank	1800
Smart Watch	6500
Headphones	4200
Tablet	22000
Web Camera	2800
PRODUCT_NAME PRICE	
External Hard Disk	5200
Gaming Monitor	18500
Laptop Stand	1500
Wireless Earbuds	3500

15 rows selected.

4. Display all orders placed by customers.

```
SQL> SELECT CONCAT(c.First_Name,c.Last_Name) AS Customer_Name,
2 o.OrderID, o.TotalAmount,o.OrderStatus
3 FROM Customer c JOIN Orders o
4 ON c.Customer_ID = o.CustomerID;
```

CUSTOMER_NAME	ORDERID	TOTALAMOUNT	ORDERSTATUS
ArjunKumar	ORD0001	55850	Delivered
PriyaSharma	ORD0002	3350	Delivered
RahulMenon	ORD0003	28000	Delivered
AnanyaIyer	ORD0004	8300	Delivered
VikramReddy	ORD0005	22000	Processing
MeeraKrishnan	ORD0006	7000	Delivered
SureshNair	ORD0007	6500	Shipped
KavyaRajan	ORD0008	5700	Delivered
KarthikSubramanian	ORD0009	18500	Processing
DivyaPillai	ORD0010	5200	Delivered
AdityaVerma	ORD0011	7000	Shipped
CUSTOMER_NAME	ORDERID	TOTALAMOUNT	ORDERSTATUS
SnehaPatel	ORD0012	2450	Delivered
LakshmiVenkatesan	ORD0014	28000	Delivered
NithinMurugan	ORD0015	8350	Processing

14 rows selected.

5. Retrieve payment details.

```
SQL> SELECT * FROM Payment;
```

PAYMENT	ORDERID	PAYMENTDA	PAYMENTMODE	PAYMENTSTA	TRANSACTIONAMOUNT
PAY0001	ORD0001	10-JAN-26	Credit Card	Completed	55850
PAY0002	ORD0002	25-JAN-26	UPI	Completed	3350
PAY0003	ORD0004	18-FEB-26	Debit Card	Completed	8300
PAY0004	ORD0006	15-MAR-26	Net Banking	Completed	7000
PAY0006	ORD0010	18-MAY-26	UPI	Completed	5200
PAY0007	ORD0012	20-JUN-26	Credit Card	Completed	2450
PAY0008	ORD0014	15-AUG-26	Debit Card	Completed	28000
PAY0009	ORD0014	14-AUG-26	Debit Card	Failed	28000
PAY0005	ORD0008	20-APR-26	COD	Completed	5700

9 rows selected.

2. Apply Filtering Conditions Using WHERE

1. Find products with price greater than ₹5000.

```
SQL> SELECT * FROM Product WHERE Price > 5000;
```

PRODUCT	PRODUCT_NAME	CATEGOR	PRICE
STOCK_QUANTITY			
P000001	Laptop	CAT001	55000
	25		
P000005	Smartphone	CAT004	28000
	35		
P000008	Smart Watch	CAT006	6500
	40		
PRODUCT	PRODUCT_NAME	CATEGOR	PRICE
STOCK_QUANTITY			
P000010	Tablet	CAT004	22000
	30		
P000012	External Hard Disk	CAT008	5200
	35		
P000013	Gaming Monitor	CAT009	18500
	20		

6 rows selected.

2. Display products available in stock.

```
SQL> SELECT * FROM Inventory WHERE Stock_Status = 'Available';
```

INVENTO	PRODUCT	SELLER	STOCK_QUANTITY	STOCK_STATU	LAST_UPDA
INV0001	P000001	SEL0001	25	Available	15-JAN-26
INV0002	P000002	SEL0002	100	Available	20-JAN-26
INV0003	P000003	SEL0002	60	Available	25-JAN-26
INV0004	P000004	SEL0003	45	Available	05-FEB-26
INV0005	P000005	SEL0004	35	Available	15-FEB-26
INV0006	P000006	SEL0005	150	Available	01-MAR-26
INV0007	P000007	SEL0005	75	Available	10-MAR-26
INV0008	P000008	SEL0006	40	Available	20-MAR-26
INV0009	P000009	SEL0003	55	Available	05-APR-26
INV0010	P000010	SEL0004	30	Available	15-APR-26
INV0011	P000011	SEL0007	50	Available	01-MAY-26
INV0012	P000012	SEL0008	35	Available	15-MAY-26
INV0013	P000013	SEL0009	20	Available	10-JUN-26
INV0014	P000014	SEL0010	80	Available	15-JUL-26
INV0015	P000015	SEL0003	65	Available	20-SEP-26

15 rows selected.

3. Display completed orders.

```
SQL> SELECT * FROM Orders WHERE OrderStatus = 'Delivered';
```

ORDERID	CUSTOMERID	ORDERDATE	TOTALAMOUNT	ORDERSTATUS
ORD0001	CUST000001	10-JAN-26	55850	Delivered
ORD0002	CUST000002	25-JAN-26	3350	Delivered
ORD0003	CUST000003	05-FEB-26	28000	Delivered
ORD0004	CUST000004	18-FEB-26	8300	Delivered
ORD0006	CUST000006	15-MAR-26	7000	Delivered
ORD0008	CUST000008	20-APR-26	5700	Delivered
ORD0010	CUST000010	18-MAY-26	5200	Delivered
ORD0012	CUST000012	20-JUN-26	2450	Delivered
ORD0014	CUST000014	15-AUG-26	28000	Delivered

9 rows selected.

4. Find products with rating above 4.

```
SQL> SELECT
2  P.PRODUCT_NAME,
3  ROUND(AVG(R.RATING), 2) AS AVERAGE_RATING
4  FROM PRODUCT P
5  JOIN REVIEW R
6  ON P.PRODUCT_ID = R.PRODUCTID
7  GROUP BY P.PRODUCT_ID, P.PRODUCT_NAME
8  HAVING AVG(R.RATING) > 4
9  ORDER BY AVERAGE_RATING DESC;
```

PRODUCT_NAME	AVERAGE_RATING
Smartphone	5
Bluetooth Speaker	5
Laptop	5
External Hard Disk	4.5
Headphones	4.5
Wireless Earbuds	4.5
Smart Watch	4.5

7 rows selected.

3. Sort Data Using ORDER BY

Tasks:

1. Display products from lowest to highest price.

```
SQL> SELECT * FROM Product ORDER BY Price;
```

PRODUCT	PRODUCT_NAME	CATEGOR	PRICE
STOCK_QUANTITY			
P000006	USB Cable 150	CAT005	450
P000002	Wireless Mouse 100	CAT002	850
P000014	Laptop Stand 80	CAT010	1500
PRODUCT	PRODUCT_NAME	CATEGOR	PRICE
STOCK_QUANTITY			
P000007	Power Bank 75	CAT005	1800
P000003	Mechanical Keyboard 60	CAT002	2500
P000011	Web Camera 50	CAT007	2800
PRODUCT	PRODUCT_NAME	CATEGOR	PRICE
STOCK_QUANTITY			
P000004	Bluetooth Speaker 45	CAT003	3200
P000015	Wireless Earbuds 65	CAT003	3500
P000009	Headphones 55	CAT003	4200
PRODUCT	PRODUCT_NAME	CATEGOR	PRICE
STOCK_QUANTITY			
P000012	External Hard Disk 35	CAT008	5200
P000008	Smart Watch 40	CAT006	6500
P000013	Gaming Monitor 20	CAT009	18500
PRODUCT	PRODUCT_NAME	CATEGOR	PRICE
STOCK_QUANTITY			
P000010	Tablet 30	CAT004	22000
P000005	Smartphone 35	CAT004	28000
P000001	Laptop 25	CAT001	55000
15 rows selected.			

2. Display customers alphabetically.

```
SQL> SELECT * FROM Customer ORDER BY First_Name;
```

CUSTOMER_I	FIRST_NAME	LAST_NAME	PHONE	CREATED_A
CUST000011	Aditya	Verma	9876501234	14-DEC-25
	aditya.verma@gmail.com			
CUST000004	Ananya	Iyer	9876123450	22-AUG-24
	ananya.iyer@gmail.com			
CUST000001	Arjun	Kumar	9876543210	15-JAN-24
	arjun.kumar@gmail.com			
CUST000010	Divya	Pillai	9098765432	30-SEP-25
	divya.pillai@gmail.com			
CUST000009	Karthik	Subramanian	8899776655	09-JUL-25
	karthik.subramanian@gmail.com			
CUST000008	Kavya	Rajan	9789012345	25-MAY-25
	kavya.rajan@gmail.com			
CUST000014	Lakshmi	Venkatesan	9543210987	08-JUL-26
	lakshmi.venkatesan@gmail.com			
CUST000006	Meera	Krishnan	9345678120	18-JAN-25
	meera.krishnan@gmail.com			
CUST000015	Nithin	Murugan	9432109876	03-OCT-26
	nithin.murugan@gmail.com			
CUST000002	Priya	Sharma	9123456780	28-MAR-24
	priya.sharma@gmail.com			
CUST000003	Rahul	Menon	9988776655	10-JUN-24
	rahul.menon@gmail.com			
CUST000013	Rohan	Bose	9654321098	17-APR-26
	rohan.bose@gmail.com			
CUST000012	Sneha	Patel	9765432109	20-FEB-26
	sneha.patel@gmail.com			
CUST000007	Suresh	Nair	9567891234	12-MAR-25
	suresh.nair@gmail.com			
CUST000005	Vikram	Reddy	9012345678	05-NOV-24
	vikram.reddy@gmail.com			

15 rows selected.

- Find top expensive products.

```
SQL> SELECT * FROM Product ORDER BY Price DESC
2 FETCH FIRST 5 ROWS ONLY;
```

PRODUCT	PRODUCT_NAME	CATEGOR	PRICE
STOCK_QUANTITY			
P000001	Laptop 25	CAT001	55000
P000005	Smartphone 35	CAT004	28000
P000010	Tablet 30	CAT004	22000
P000013	Gaming Monitor 20	CAT009	18500
P000008	Smart Watch 40	CAT006	6500

- Display latest orders first.

```
SQL> SELECT * FROM Orders ORDER BY OrderDate DESC;
```

ORDERID	CUSTOMERID	ORDERDATE	TOTALAMOUNT	ORDERSTATUS
ORD0015	CUST000015	25-SEP-26	8350	Processing
ORD0014	CUST000014	15-AUG-26	28000	Delivered
ORD0012	CUST000012	20-JUN-26	2450	Delivered
ORD0011	CUST000011	03-JUN-26	7000	Shipped
ORD0010	CUST000010	18-MAY-26	5200	Delivered
ORD0009	CUST000009	05-MAY-26	18500	Processing
ORD0008	CUST000008	20-APR-26	5700	Delivered
ORD0007	CUST000007	01-APR-26	6500	Shipped
ORD0006	CUST000006	15-MAR-26	7000	Delivered
ORD0005	CUST000005	02-MAR-26	22000	Processing
ORD0004	CUST000004	18-FEB-26	8300	Delivered
ORD0003	CUST000003	05-FEB-26	28000	Delivered
ORD0002	CUST000002	25-JAN-26	3350	Delivered
ORD0001	CUST000001	10-JAN-26	55850	Delivered

14 rows selected.

4. Retrieve Unique Values Using DISTINCT

Tasks:

- Display unique product categories.

CATEGORY
Audio Devices
Clothing
Computer Accessories
Electronics
Home
Jewellery
Laptop Accessories
Laptops
Mobile Devices
Monitors
Pharmacy
Power Accessories
Storage Devices
Wearables
Web Accessories

15 rows selected.

2. Find different payment methods used by customers.

```
SQL> SELECT DISTINCT PaymentMode AS Payment_Mode FROM Payment;

PAYMENT_MODE
-----
Credit Card
UPI
Debit Card
Net Banking
COD
```

5. Search Products Based on Conditions

Tasks:

Write SQL queries to:

1. Find products within a specific price range. (Product Between 20,000 to 40,000)

```
SQL> SELECT * FROM PRODUCT WHERE Price BETWEEN 20000 AND 40000;

PRODUCT  PRODUCT_NAME                                CATEGOR  PRICE
-----
STOCK_QUANTITY
P000005 Smartphone                          CAT004    28000
      35
P000010 Tablet                              CAT004    22000
      30
```

2. Display products belonging to a particular category.

```
SQL> SELECT * FROM Product WHERE Category_ID = 'CAT005';

PRODUCT  PRODUCT_NAME                                CATEGOR  PRICE
-----
STOCK_QUANTITY
P000006 USB Cable                          CAT005     450
      150
P000007 Power Bank                        CAT005    1800
      75
```

3. Find products currently available in inventory.

```
SQL> SELECT * FROM Inventory WHERE Stock_Status = 'Available';

INVENTO  PRODUCT  SELLER_  STOCK_QUANTITY  STOCK_STATU  LAST_UPDA
-----
INV0001 P000001 SEL0001      25 Available  15-JAN-26
INV0002 P000002 SEL0002     100 Available  20-JAN-26
INV0003 P000003 SEL0002      60 Available  25-JAN-26
INV0004 P000004 SEL0003      45 Available  05-FEB-26
INV0005 P000005 SEL0004      35 Available  15-FEB-26
INV0006 P000006 SEL0005     150 Available  01-MAR-26
INV0007 P000007 SEL0005      75 Available  10-MAR-26
INV0008 P000008 SEL0006      40 Available  20-MAR-26
INV0009 P000009 SEL0003      55 Available  05-APR-26
INV0010 P000010 SEL0004      30 Available  15-APR-26
INV0011 P000011 SEL0007      50 Available  01-MAY-26

INVENTO  PRODUCT  SELLER_  STOCK_QUANTITY  STOCK_STATU  LAST_UPDA
-----
INV0012 P000012 SEL0008      35 Available  15-MAY-26
INV0013 P000013 SEL0009      20 Available  10-JUN-26
INV0014 P000014 SEL0010      80 Available  15-JUL-26
INV0015 P000015 SEL0003      65 Available  20-SEP-26

15 rows selected.
```

4. Search products using product name.

```
SQL> SELECT * FROM Product WHERE Product_Name = 'Power Bank';
```

PRODUCT	PRODUCT_NAME	CATEGOR	PRICE
STOCK_QUANTITY			
P0000007	Power Bank	CAT005	1800
	75		

5. Find low-stock products.

```
SQL> SELECT * FROM Inventory WHERE Stock_Quantity < 30;
```

INVENTO	PRODUCT	SELLER_	STOCK_QUANTITY	STOCK_STATU	LAST_UPDA
INV0001	P000001	SEL0001	25	Available	15-JAN-26
INV0013	P000013	SEL0009	20	Available	10-JUN-26

6. Retrieve Customer and Product Information

Tasks:

1. Display customer details with their orders.

```

SQL> SELECT
2      C.CUSTOMER_ID
3      C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
4      C.EMAIL,
5      C.PHONE,
6      O.ORDERID AS ORDER_ID,
7      O.ORDERDATE AS ORDER_DATE,
8      O.TOTALAMOUNT AS TOTAL_AMOUNT,
9      O.ORDERSTATUS AS ORDER_STATUS
10 FROM CUSTOMER C
11 JOIN ORDERS O
12   ON C.CUSTOMER_ID = O.CUSTOMERID
13 ORDER BY C.CUSTOMER_ID, O.ORDERDATE;

```

CUSTOMER_I	CUSTOMER_NAME	PHONE	ORDER_I	ORDER_DAT	TOTAL_AMOUNT
ORDER_STATUS					
CUST000001	Arjun Kumar arjun.kumar@gmail.com Delivered	9876543210	ORD0001	10-JAN-26	55850
CUST000002	Priya Sharma priya.sharma@gmail.com Delivered	9123456780	ORD0002	25-JAN-26	3350
CUSTOMER_I CUSTOMER_NAME					
EMAIL	PHONE	ORDER_I	ORDER_DAT	TOTAL_AMOUNT	
ORDER_STATUS					
CUST000003	Rahul Menon rahul.menon@gmail.com Delivered	9988776655	ORD0003	05-FEB-26	28000
CUST000004	Ananya Iyer ananya.iyer@gmail.com	9876123450	ORD0004	18-FEB-26	8300
CUSTOMER_I CUSTOMER_NAME					
EMAIL	PHONE	ORDER_I	ORDER_DAT	TOTAL_AMOUNT	
ORDER_STATUS					
Delivered					
CUST000005	Vikram Reddy vikram.reddy@gmail.com Processing	9012345678	ORD0005	02-MAR-26	22000
CUST000006	Meera Krishnan				
CUSTOMER_I CUSTOMER_NAME					
EMAIL	PHONE	ORDER_I	ORDER_DAT	TOTAL_AMOUNT	
ORDER_STATUS					
meera.krishnan@gmail.com	9345678120	ORD0006	15-MAR-26	7000	
Delivered					
CUST000007	Suresh Nair suresh.nair@gmail.com Shipped	9567891234	ORD0007	01-APR-26	6500

CUSTOMER_I		CUSTOMER_NAME				
EMAIL	PHONE	ORDER_I	ORDER_DAT	TOTAL_AMOUNT		
ORDER_STATUS						
CUST000008	Kavya Rajan					
kavya.rajan@gmail.com	9789012345	ORD0008	20-APR-26	5700		
Delivered						
CUST000009	Karthik Subramanian					
karthik.subramanian@gmail.com	8899776655	ORD0009	05-MAY-26	18500		
Processing						
CUSTOMER_I		CUSTOMER_NAME				
EMAIL	PHONE	ORDER_I	ORDER_DAT	TOTAL_AMOUNT		
ORDER_STATUS						
CUST000010	Divya Pillai					
divya.pillai@gmail.com	9098765432	ORD0010	18-MAY-26	5200		
Delivered						
CUST000011	Aditya Verma					
aditya.verma@gmail.com	9876501234	ORD0011	03-JUN-26	7000		
CUSTOMER_I		CUSTOMER_NAME				
EMAIL	PHONE	ORDER_I	ORDER_DAT	TOTAL_AMOUNT		
ORDER_STATUS						
Shipped						
CUST000012	Sneha Patel					
sneha.patel@gmail.com	9765432109	ORD0012	20-JUN-26	2450		
Delivered						
CUST000014	Lakshmi Venkatesan					
CUSTOMER_I		CUSTOMER_NAME				
EMAIL	PHONE	ORDER_I	ORDER_DAT	TOTAL_AMOUNT		
ORDER_STATUS						
lakshmi.venkatesan@gmail.com	9543210987	ORD0014	15-AUG-26	28000		
Delivered						
CUST000015	Nithin Murugan					
nithin.murugan@gmail.com	9432109876	ORD0015	25-SEP-26	8350		
Processing						
14 rows selected.						

2. Display product details with category information.

```
SQL> SELECT
2     P.PRODUCT_ID,
3     P.PRODUCT_NAME,
4     P.PRICE,
5     P.STOCK_QUANTITY,
6     C.CATEGORY_ID,
7     C.CATEGORY_NAME,
8     C.DESCRPTION
9 FROM PRODUCT P
10 JOIN CATEGORY C
11 ON P.CATEGORY_ID = C.CATEGORY_ID
12 ORDER BY C.CATEGORY_NAME, P.PRODUCT_NAME;
```

PRODUCT	PRODUCT_NAME	PRICE
STOCK_QUANTITY	CATEGOR	CATEGORY_NAME
DESCRIPTION		
P000004	Bluetooth Speaker 45 CAT003 Audio Devices Speakers headphones and earbuds	3200
P000009	Headphones 55 CAT003 Audio Devices Speakers headphones and earbuds	4200
PRODUCT	PRODUCT_NAME	PRICE
STOCK_QUANTITY	CATEGOR	CATEGORY_NAME
DESCRIPTION		
P000015	Wireless Earbuds 65 CAT003 Audio Devices Speakers headphones and earbuds	3500
P000003	Mechanical Keyboard 60 CAT002 Computer Accessories	2500
PRODUCT	PRODUCT_NAME	PRICE
STOCK_QUANTITY	CATEGOR	CATEGORY_NAME
DESCRIPTION		
	Mouse and keyboard accessories	
P000002	Wireless Mouse 100 CAT002 Computer Accessories Mouse and keyboard accessories	850
P000014	Laptop Stand	1500
PRODUCT	PRODUCT_NAME	PRICE
STOCK_QUANTITY	CATEGOR	CATEGORY_NAME
DESCRIPTION		
	80 CAT010 Laptop Accessories Laptop stands and accessories	
P000001	Laptop 25 CAT001 Laptops Laptop computers and notebooks	55000

PRODUCT	PRODUCT_NAME	PRICE
STOCK_QUANTITY	CATEGOR	CATEGORY_NAME
DESCRIPTION		
P000005	Smartphone 35 CAT004 Mobile Devices Smartphones and tablets	28000
P000010	Tablet 30 CAT004 Mobile Devices Smartphones and tablets	22000
PRODUCT	PRODUCT_NAME	PRICE
STOCK_QUANTITY	CATEGOR	CATEGORY_NAME
DESCRIPTION		
P000013	Gaming Monitor 20 CAT009 Monitors Gaming and computer monitors	18500
P000007	Power Bank 75 CAT005 Power Accessories	1800
PRODUCT	PRODUCT_NAME	PRICE
STOCK_QUANTITY	CATEGOR	CATEGORY_NAME
DESCRIPTION		
	USB cables and power banks	
P000006	USB Cable 150 CAT005 Power Accessories USB cables and power banks	450
P000012	External Hard Disk	5200
PRODUCT	PRODUCT_NAME	PRICE
STOCK_QUANTITY	CATEGOR	CATEGORY_NAME
DESCRIPTION		
	35 CAT008 Storage Devices External hard disks and storage	
P000008	Smart Watch 40 CAT006 Wearables Smart watches and wearable devices	6500
PRODUCT	PRODUCT_NAME	PRICE
STOCK_QUANTITY	CATEGOR	CATEGORY_NAME
DESCRIPTION		
P000011	Web Camera 50 CAT007 Web Accessories Web cameras and related devices	2800

15 rows selected.

3. Find customers who purchased a specific product.

```
SQL> SELECT DISTINCT
2      C.CUSTOMER_ID,
3      C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
4      C.EMAIL,
5      C.PHONE
6 FROM CUSTOMER C
7 JOIN ORDERS O
8      ON C.CUSTOMER_ID = O.CUSTOMERID
9 JOIN ORDERDETAIL OD
10      ON O.ORDERID = OD.ORDERID
11 JOIN PRODUCT P
12      ON OD.PRODUCTID = P.PRODUCT_ID
13 WHERE P.PRODUCT_ID = 'P000009';

CUSTOMER_I CUSTOMER_NAME
-----
EMAIL
-----
PHONE
-----
CUST000006 Meera Krishnan
meera.krishnan@gmail.com      9345678120

CUST000011 Aditya Verma
aditya.verma@gmail.com        9876501234
```

4. Display products purchased by each customer.

```
SQL> SELECT
2      C.CUSTOMER_ID,
3      C.FIRST_NAME || ' ' || C.LAST_NAME AS CUSTOMER_NAME,
4      O.ORDERID AS ORDER_ID,
5      P.PRODUCT_NAME,
6      OD.QUANTITY,
7      OD.PRICE,
8      OD.QUANTITY * OD.PRICE AS LINE_AMOUNT
9 FROM CUSTOMER C
10 JOIN ORDERS O
11      ON C.CUSTOMER_ID = O.CUSTOMERID
12 JOIN ORDERDETAIL OD
13      ON O.ORDERID = OD.ORDERID
14 JOIN PRODUCT P
15      ON OD.PRODUCTID = P.PRODUCT_ID
16 ORDER BY C.CUSTOMER_ID, O.ORDERDATE, P.PRODUCT_NAME;

CUSTOMER_I CUSTOMER_NAME
-----
ORDER_I
-----
PRODUCT_NAME
-----
QUANTITY
-----
PRICE
-----
LINE_AMOUNT
-----
CUST000001 Arjun Kumar
Laptop
55000
ORD00001
1
55000
CUST000001 Arjun Kumar
Wireless Mouse
850
ORD00001
1
850
CUSTOMER_I CUSTOMER_NAME
-----
ORDER_I
-----
PRODUCT_NAME
-----
QUANTITY
-----
PRICE
-----
LINE_AMOUNT
-----
CUST000002 Priya Sharma
Bluetooth Speaker
3200
ORD00002
1
3200
CUST000002 Priya Sharma
USB Cable
450
ORD00002
1
450
CUSTOMER_I CUSTOMER_NAME
-----
ORDER_I
-----
PRODUCT_NAME
-----
QUANTITY
-----
PRICE
-----
LINE_AMOUNT
-----
CUST000003 Rahul Menon
Smartphone
168000
ORD00003
6
28000
CUST000004 Ananya Iyer
Smart Watch
6500
ORD00004
1
6500
CUST000004 Ananya Iyer
USB Cable
1800
ORD00004
4
450
```

CUSTOMER_I	CUSTOMER_NAME	ORDER_I	
PRODUCT_NAME		QUANTITY	PRICE
LINE_AMOUNT			
CUST000005	Vikram Reddy	ORD0005	
Tablet		1	22000
22000			
CUST000006	Meera Krishnan	ORD0006	
Headphones		1	4200
4200			
CUSTOMER_I	CUSTOMER_NAME	ORDER_I	
PRODUCT_NAME		QUANTITY	PRICE
LINE_AMOUNT			
CUST000006	Meera Krishnan	ORD0006	
Wireless Earbuds		1	3500
3500			
CUST000007	Suresh Nair	ORD0007	
Smart Watch		1	6500
CUSTOMER_I	CUSTOMER_NAME	ORDER_I	
PRODUCT_NAME		QUANTITY	PRICE
LINE_AMOUNT			
			6500
CUST000008	Kavya Rajan	ORD0008	
External Hard Disk		1	5200
5200			
CUST000008	Kavya Rajan	ORD0008	
CUSTOMER_I	CUSTOMER_NAME	ORDER_I	
PRODUCT_NAME		QUANTITY	PRICE
LINE_AMOUNT			
USB Cable		1	450
450			
CUST000009	Karthik Subramanian	ORD0009	
Gaming Monitor		1	18500
18500			
CUSTOMER_I	CUSTOMER_NAME	ORDER_I	
PRODUCT_NAME		QUANTITY	PRICE
LINE_AMOUNT			
CUST000010	Diya Pillai	ORD0010	
External Hard Disk		1	5200
5200			
CUST000011	Aditya Verma	ORD0011	
Headphones		1	4200
4200			

CUSTOMER_I	CUSTOMER_NAME	ORDER_I	
PRODUCT_NAME		QUANTITY	PRICE
LINE_AMOUNT			
CUST000011	Aditya Verma	ORD0011	
Wireless Earbuds		1	3500
3500			
CUST000012	Sneha Patel	ORD0012	
Laptop Stand		1	1500
CUSTOMER_I	CUSTOMER_NAME	ORDER_I	
PRODUCT_NAME		QUANTITY	PRICE
LINE_AMOUNT			
			1500
CUST000012	Sneha Patel	ORD0012	
Mechanical Keyboard		1	2500
2500			
CUST000015	Nithin Murugan	ORD0015	
CUSTOMER_I	CUSTOMER_NAME	ORDER_I	
PRODUCT_NAME		QUANTITY	PRICE
LINE_AMOUNT			
Wireless Earbuds		1	3500
3500			

20 rows selected.

7. Apply Multiple Filtering Conditions

Tasks:

- Find electronics products costing more than ₹10000.

```
SQL> SELECT
  2     P.PRODUCT_ID,
  3     P.PRODUCT_NAME,
  4     C.CATEGORY_NAME,
  5     P.PRICE
  6 FROM PRODUCT P
  7 JOIN CATEGORY C
  8     ON P.CATEGORY_ID = C.CATEGORY_ID
  9 WHERE P.PRICE > 10000
 10 ORDER BY P.PRICE DESC;
```

PRODUCT	PRODUCT_NAME	PRICE
P000001	Laptop	55000
P000005	Smartphone	28000
P000010	Tablet	22000
P000013	Gaming Monitor	18500

2. Search products containing the word "Mobile".

```
SQL> SELECT
  2     PRODUCT_ID,
  3     PRODUCT_NAME,
  4     PRICE,
  5     STOCK_QUANTITY
  6 FROM PRODUCT
  7 WHERE UPPER(PRODUCT_NAME) LIKE '%MOBILE%';

no rows selected
```

3. Find orders placed within a specific date range.

```
SQL> SELECT
  2     ORDERID,
  3     CUSTOMERID,
  4     ORDERDATE,
  5     TOTALAMOUNT,
  6     ORDERSTATUS
  7 FROM ORDERS
  8 WHERE ORDERDATE BETWEEN DATE '2026-03-01' AND DATE '2026-06-30'
  9 ORDER BY ORDERDATE;
```

ORDERID	CUSTOMERID	ORDERDATE	TOTALAMOUNT	ORDERSTATUS
ORD0005	CUST000005	02-MAR-26	22000	Processing
ORD0006	CUST000006	15-MAR-26	7000	Delivered
ORD0007	CUST000007	01-APR-26	6500	Shipped
ORD0008	CUST000008	20-APR-26	5700	Delivered
ORD0009	CUST000009	05-MAY-26	18500	Processing
ORD0010	CUST000010	18-MAY-26	5200	Delivered
ORD0011	CUST000011	03-JUN-26	7000	Shipped
ORD0012	CUST000012	20-JUN-26	2450	Delivered

8 rows selected.

8. Generate Basic Business Reports

Report 1: Product Availability Report

Display:

- Product name

- Price
- Stock quantity
- Availability status

```
SQL> SELECT
2     PRODUCT_NAME,
3     PRICE,
4     STOCK_QUANTITY,
5     CASE
6         WHEN STOCK_QUANTITY > 0 THEN 'Available'
7         ELSE 'Out of Stock'
8     END AS AVAILABILITY_STATUS
9 FROM PRODUCT
10 ORDER BY PRODUCT_NAME;
```

PRODUCT_NAME	PRICE	STOCK_QUANTITY
AVAILABILITY		
Bluetooth Speaker Available	3200	45
External Hard Disk Available	5200	35
Gaming Monitor Available	18500	20

PRODUCT_NAME	PRICE	STOCK_QUANTITY
AVAILABILITY		
Headphones Available	4200	55
Laptop Available	55000	25
Laptop Stand Available	1500	80

PRODUCT_NAME	PRICE	STOCK_QUANTITY
AVAILABILITY		
Mechanical Keyboard Available	2500	60
Power Bank Available	1800	75
Smart Watch Available	6500	40

PRODUCT_NAME	PRICE	STOCK_QUANTITY
AVAILABILITY		
Smartphone Available	28000	35
Tablet Available	22000	30
USB Cable Available	450	150

PRODUCT_NAME	PRICE	STOCK_QUANTITY
AVAILABILITY		
Web Camera Available	2800	50
Wireless Earbuds Available	3500	65
Wireless Mouse Available	850	100

15 rows selected.

Report 2: Customer Report

Generate:

- Total number of customers

```
SQL> SELECT COUNT(*) AS TOTAL_CUSTOMERS
2 FROM CUSTOMER;

TOTAL_CUSTOMERS
-----
15
```

- New customer registrations

```
SQL> SELECT
2 CUSTOMER_ID
3 FIRST_NAME || ' ' || LAST_NAME AS CUSTOMER_NAME,
4 EMAIL,
5 CREATED_AT
6 FROM CUSTOMER
7 WHERE CREATED_AT >= DATE '2026-01-01'
8 AND CREATED_AT < DATE '2027-01-01'
9 ORDER BY CREATED_AT DESC;

CUSTOMER_I CUSTOMER_NAME
-----
EMAIL CREATED_A
-----
CUST000015 Nithin Murugan
nithin.murugan@gmail.com 03-OCT-26

CUST000014 Lakshmi Venkatesan
lakshmi.venkatesan@gmail.com 08-JUL-26

CUST000013 Rohan Bose
rohan.bose@gmail.com 17-APR-26

CUSTOMER_I CUSTOMER_NAME
-----
EMAIL CREATED_A
-----
CUST000012 Sneha Patel
sneha.patel@gmail.com 20-FEB-26
```

Report 3: Order Report

Display:

- Total orders

```
SQL> SELECT COUNT(*) AS TOTAL_ORDERS
2 FROM ORDERS;

TOTAL_ORDERS
-----
14
```

- Completed orders

```
SQL> SELECT COUNT(*) AS COMPLETED_ORDERS
2 FROM ORDERS
3 WHERE ORDERSTATUS = 'Delivered';

COMPLETED_ORDERS
-----
9
```

- Pending orders

```
SQL> SELECT COUNT(*) AS PENDING_ORDERS
2 FROM ORDERS
3 WHERE ORDERSTATUS IN ('Processing', 'Shipped');

PENDING_ORDERS
-----
5
```

● Cancelled orders

```
SQL> SELECT COUNT(*) AS CANCELLED_ORDERS
2 FROM ORDERS
3 WHERE ORDERSTATUS = 'Cancelled';

CANCELLED_ORDERS
-----
0
```

Report 4: Product Performance Report

Find:

● Highest-priced products

```
SQL> SELECT
2 PRODUCT_ID,
3 PRODUCT_NAME,
4 PRICE
5 FROM PRODUCT
6 ORDER BY PRICE DESC;

PRODUCT PRODUCT_NAME PRICE
-----
P000001 Laptop 55000
P000005 Smartphone 28000
P000010 Tablet 22000
P000013 Gaming Monitor 18500
P000008 Smart Watch 6500
P000012 External Hard Disk 5200
P000009 Headphones 4200
P000015 Wireless Earbuds 3500
P000004 Bluetooth Speaker 3200
P000011 Web Camera 2800
P000003 Mechanical Keyboard 2500

PRODUCT PRODUCT_NAME PRICE
-----
P000007 Power Bank 1800
P000014 Laptop Stand 1500
P000002 Wireless Mouse 850
P000006 USB Cable 450

15 rows selected.
```

● Most reviewed products

```
SQL> SELECT
2 P.PRODUCT_ID,
3 P.PRODUCT_NAME,
4 COUNT(R.REVIEWID) AS REVIEW_COUNT
5 FROM PRODUCT P
6 JOIN REVIEW R
7 ON P.PRODUCT_ID = R.PRODUCTID
8 GROUP BY P.PRODUCT_ID, P.PRODUCT_NAME
9 ORDER BY REVIEW_COUNT DESC;

PRODUCT PRODUCT_NAME REVIEW_COUNT
-----
P000006 USB Cable 3
P000008 Smart Watch 2
P000012 External Hard Disk 2
P000015 Wireless Earbuds 2
P000009 Headphones 2
P000014 Laptop Stand 1
P000010 Tablet 1
P000005 Smartphone 1
P000001 Laptop 1
P000004 Bluetooth Speaker 1
P000002 Wireless Mouse 1

11 rows selected.
```

● Available products

```
SQL> SELECT
2     PRODUCT_ID,
3     PRODUCT_NAME,
4     PRICE,
5     STOCK_QUANTITY
6 FROM PRODUCT
7 WHERE STOCK_QUANTITY > 0
8 ORDER BY STOCK_QUANTITY DESC;
```

PRODUCT	PRODUCT_NAME	PRICE
STOCK_QUANTITY		
P000006	USB Cable 150	450

P000002	Wireless Mouse 100	850
---------	-----------------------	-----

P000014	Laptop Stand 80	1500
---------	--------------------	------

PRODUCT	PRODUCT_NAME	PRICE
STOCK_QUANTITY		

P000007	Power Bank 75	1800
---------	------------------	------

P000015	Wireless Earbuds 65	3500
---------	------------------------	------

P000003	Mechanical Keyboard 60	2500
---------	---------------------------	------

PRODUCT	PRODUCT_NAME	PRICE
STOCK_QUANTITY		

P000009	Headphones 55	4200
---------	------------------	------

P000011	Web Camera 50	2800
---------	------------------	------

P000004	Bluetooth Speaker 45	3200
---------	-------------------------	------

PRODUCT	PRODUCT_NAME	PRICE
STOCK_QUANTITY		

P000008	Smart Watch 40	6500
---------	-------------------	------

P000005	Smartphone 35	28000
---------	------------------	-------

P000012	External Hard Disk 35	5200
---------	--------------------------	------

PRODUCT	PRODUCT_NAME	PRICE
STOCK_QUANTITY		

P000010	Tablet 30	22000
---------	--------------	-------

P000001	Laptop 25	55000
---------	--------------	-------

P000013	Gaming Monitor 20	18500
---------	----------------------	-------

15 rows selected.